Longitudinal and cross-sectional perspectives on elementary school children's post-error slowing in reading and arithmetic

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Abstract

This study investigated the development of primary school children's error-related adjustments in reading and arithmetic. N = 622 children from 2nd and 4th grade were tested on two tablet-based academic tasks twice in six months: an arithmetic task composed of addition and doubling items with two answer options and a silent word reading task matching words with four pictorial answer options. Children in both grades showed robust post-error slowing (PES) in both reading and arithmetic, indicating reliable monitoring and strategy use in the form of upregulating cognitive control mechanisms. Overall, post-error slowing was larger in arithmetic than in reading, independent of grade and measurement point. Fourth graders showed larger PES than 2nd graders, especially at T1, and both grades' PES increased from T1 to T2, suggesting that task experience, improving monitoring skills, and strategy discovery play an important role. In terms of the benefits of PES for performance accuracy, PES was linked to better accuracy after the error in the arithmetic task, particularly for younger children. This was not the case for reading. Results are discussed in terms of often overlooked educationally relevant strategies in arithmetic, possible strategy's utilization deficiency, developing monitoring and cognitive control skills in this age range and differing processing demands across academic domains.

Keywords: post-error slowing, reading, arithmetic, developmental progression in strategy use, elementary school children

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Children's mathematical (basic arithmetic) and reading proficiency are two of the most important academic skills. Testing children's skills in these domains is typically done by presenting several similar items and pre-defining a time in which students must complete the test. Imagine Susie, a 4th grade student, in such a situation: when the test runs smoothly, Susie will feel certain and she will most likely speed up and promptly work herself through the test items. However, when she encounters problems or makes a mistake as the test proceeds, Susie will hopefully detect her error, feel less confident and possibly slow down to avoid more mistakes. As the example illustrates, a child's ability to detect errors and adapt her task strategy is crucial for test outcomes (e.g., Bellon et al., 2020; Krebs & Roebers, 2010; Steiner et al., 2019). In the present contribution, one marker of children's error monitoring and adaptive strategy use that has often been overlooked will be explored in the context of reading and basic arithmetic: post-error slowing (PES; Botvinick, et al., 2001; Braver, 2012; de Mooij et al., 2022).

Traditionally, post-error slowing was investigated in adults, mirroring goal-oriented, monitoring based cognitive control strategies (Botvinick et al., 2001; Braver, 2012; Moser et al., 2011; Rabbitt, 1966). PES is thought to reflect a monitoring based, strategic upregulation of cognitive control processes following an error, which should – ideally – lead to more accurate responding on the next trial; that is, should ideally lead to improved post-error accuracy (PEA; Botvinick et al., 2001; Laming, 1968). Generally, post-error slowing occurs when overall task performance is high (> 75% accuracy), when trials are of similar difficulty such as in classical cognitive conflict tasks, and when performance crucially depends on the balancing of speed and accuracy (Danielsmeier & Ullsperger, 2011; Notebaert et al., 2009; Schroder et al., 2019).

Post-error slowing can also be observed in children. School-aged children substantially and reliably slow down after errors, that is, show substantial PES in various

multi-trial cognitive conflict tasks (Cravet & Ger, 2026; Ger & Buehler, 2024; Gupta et al., 2009; Rabbitt, 1966). As to the magnitude of PES when comparing different age groups cross-sectionally, there is somewhat inconsistent evidence. While some studies found an age-related decrease of PES magnitude over the elementary school years (Dubravac et al., 2022; Smulders et al., 2016), interpreted as fine-tuning and increased efficiency of PES, others reported an increase (Devervaud et al., 2020; Overbye et al., 2019), possibly indicative of improved error monitoring, or, within narrower age ranges, no age differences (Ger & Roebers, 2024a; Masina et al., 2018; Roebers, 2022a). A likely explanation for this inconsistent pattern of developmental differences is children's lack of experience with speeded cognitive conflict tasks calling, first, for the discovery of a speed-accuracy balancing strategy that, only later, becomes fine-tuned with increasing experience and age (Camerota et al., 2019; Chevalier et al., 2017; Kubota et al., 2020; Niebaum et al., 2021; Smigasiewicz, Ambrosi, Blaye, & Burle, 2020).

Unfortunately, nearly all this work on post-error slowing and children's balancing of speed and accuracy within a task relies on classic conflict paradigms using abstract stimuli and highly uniform cognitive response demands (Camerota et al., 2019; Danielmeier & Ullsperger, 2011; Dubravac et al., 2022; van der Borght et al., 2016). While these paradigms capture core monitoring and adaptation mechanisms, they lack the domain-specific processing requirements of ecologically valid academic tasks, such as reading or arithmetic. To our knowledge, only two studies have examined the development of PES in school-relevant tasks in children. Using large-scale data from a national online learning platform with mathematics (basic and advanced arithmetic) and language games for primary school children, de Mooij and colleagues (2022) documented substantial post-error slowing in nearly all tasks, with PES being overall larger in arithmetic than in language activities, and with an age-related increase between ages of 5 to 9 years (but not further). In contrast, Jacobs and colleagues (2024) assessed 2nd and 3rd graders' PES in the context of single-digit-multiplication tasks twice with a delay of one year and found some evidence of substantial post-error speed adjustments but no developmental changes in PES over time.

Given the sparce and inconsistent evidence of PES in elementary children's academic tasks including cross-sectional and longitudinal changes, and given that PES might be a good marker for a monitoring based strategy use in speeded multi-trial tasks, our study aimed to fill this gap by including 2nd and 4th graders, following them up over 6 month, and assessing PES in the context of simple word reading and basic arithmetic (addition and doubling).

Importantly, the putative function of post-error slowing is a reliable increase in responses' accuracy (post-error accuracy; PEA), and in adult samples, this is typically found (Botvinick et al., 2001; Laming, 1968). In children, again, the evidence is poor and mixed. While in some developmental studies, PES was linked to post-error accuracy (Overbye et al., 2019; Pietto et al., 2023), it was not the case in others (Arnett et al., 2022; Torpey et al., 2012). This may not be surprising given that children have repeatedly been found to use a certain strategy (for example, in a memory task), but their performance does not benefit from it (Bjorklund & Coyle, 1995; Miller & Seier, 1994). Investigated under the term "utilization deficiency" (Bjorklund et al., 1997; Miller et al., 1994), it refers to a developmental phenomenon occurring in many different task types because early strategy use is effortful and resource-demanding, rendering it inefficient. Children overcome utilization deficiencies through practice, advances in metacognitive monitoring (error detection), and task-specific experience (Chevalier et al., 2015; Schneider & Löffler, 2016; Selmeczy, Ghetti, Zheng, Porter, & Trzesniewski, 2021). After a transition phase, performance typically benefits from monitoring and strategy use (for a review see Roebers, 2022b). Interestingly, while de Mooij et al. (2022) found PES to be substantially related to PEA, Jacobs et al. (2024) did not observe this link, although they reported significant PEA. This underscores the potential relevance of PES for academic achievement, albeit with some variability across studies.

The Present Study

Two major unresolved questions were addressed in the present contribution: First, both studies provide only limited insight into cross-sectional and longitudinal developments of PES in the context of curriculum-based skills in elementary school, such as reading and arithmetic. We therefore tackled simple word reading and basic arithmetic. Shedding light on

children's strategy to optimize speed and accuracy in timed tasks across age groups and over time allows testing whether monitoring-based strategies identified in laboratory paradigms generalize to ecologically valid academic contexts. Second, our understanding of PES as a strategy to optimize performance (in terms of PEA) in academic contexts is very limited. Insights into the role of post-error slowing for performance are both of theoretical and applied relevance: they provide insights into possible monitoring-based mechanisms underlying individual differences in academic performance beyond domain-specific skills (e.g., Bellon et al., 2019; Theobald et al., 2024). And it informs innovative educational approaches targeting children's error monitoring and adaptive strategy use (Brod et al., 2020; Clerc et al., 2021).

We therefore adopted a mixed cross-sectional and longitudinal design that combines within-person change over six months with between-age comparisons of 2nd and 4th graders. This age and time range coincides with a phase of schooling in which children typically show substantial gains in strategy use, metacognitive monitoring, and in core academic skills, including reading fluency (automated word recognition) and basic arithmetic fact retrieval (Geary, 2011; Roebers, 2017; 2022b; Roembke et al., 2019; Schwanenflugel et al., 2006; van der Ven et al., 2012), rendering these tasks well suited to investigate PES and PEA (Ger & Roebers, 2023a; Smulders et al., 2016; Schroder et al., 2019). The inclusion of both 2nd and 4th graders allowed us to capture broader cross-sectional age-related differences across this critical range. Moreover, the repeated measurement of PES and PEA enabled the investigation of a utilization deficiency and its possible overcoming.

We expected children to show reliable post-error slowing (PES) in both simple word reading and basic arithmetic, reflecting engagement of monitoring-based performance regulation following errors (H1). Given developmental improvements in metacognitive monitoring, strategic behavior, and efficiency of cognitive control, PES was expected to be more pronounced in 4th graders compared to 2nd graders (H2). Furthermore, PES was expected to increase longitudinally across the school year, as growing task experience and strategy automatization should promote more consistent speed–accuracy regulation (H3). Regarding the functional consequences of PES, we drew on the utilization deficiency

literature. Accordingly, PES was not expected to be associated with improved post-error accuracy (PEA) at the first measurement, but to emerge as performance-beneficial at the second measurement, reflecting increasing efficiency of monitoring-based strategy use (H4). Finally, reading and arithmetic were examined as two foundational academic domains that differ in processing demands and automatization trajectories (Geary, 2011; Schwanenflugel et al., 2006). Given the limited and inconsistent prior evidence, domain differences in PES and its relation to PEA were explored without directional hypotheses.

Method

The data for the present study were drawn from a longitudinal research project. Data collection took place during the 2024/2025 school year across two waves comprising three measurement occasions (see Participants and Procedure for details). The study protocol was approved by the Ethics Committee of the Faculty of Human Sciences of the University of Bern. All data were collected and analyzed in pseudonymized form.

Participants

The study was carried out in primary schools in the larger Bern area in Switzerland and included children from 2nd and 4th grade classes. At the first time point (T1; October–December 2024), consent was obtained for N = 704 children. Due to absences or technical issues on the test day, n = 662 completed the reading task, and n = 661 children completed the arithmetic task. At the second time point (T2; April–June 2025), three additional consents were obtained, bringing the total to N = 707 children; N = 671 children completed each task. The third assessment (T3) was scheduled approximately one week after T2, within the same April–June 2025 period.

For the present analyses, we excluded 10 third graders who were tested unintentionally in mixed 3rd/4th grade classes, as they were outside the target grades. We then focused on children who completed the reading and/or arithmetic tasks at both T1 and T2, resulting in a final sample of N = 622 for reading, and N = 620 children for mathematics. After further performance-based exclusions as described in the “Data Screening and Exclusions” section below, the final analytic samples comprised N = 614 children for the

reading task, and N = 581 children for the mathematic task. Specifically, the reading sample comprised n = 328 2nd graders (48.2% female; M age = 7.96 years, SD = 0.43) and n = 289 4th graders (45.7% female; M age = 10.02 years, SD = 0.46), while the arithmetic sample included n = 301 2nd graders (47.5% female; M age = 7.96 years, SD = 0.41) and n = 280 4th graders (45% female; M age = 10.02 years, SD = 0.45).

Classes were recruited by directly contacting school principals and class teachers. Participation was voluntary and required written informed parental consent. Children were informed that they could discontinue participation at any time without any negative consequences and without giving a reason. Pilot testing of the tasks and procedure was conducted in additional classes before the first (September 2024) and second (March 2025) measurement occasions to check feasibility and to familiarize experimenters with the procedures; these pilot data were not included in the analyses. No piloting preceded T3.

Procedure

At each measurement occasion, testing took place during regular lesson time in available classrooms at the participating schools. Sessions lasted approximately one 45-minute lesson. Children were tested in small groups using Android tablet computers (e.g., Samsung Galaxy® A7, S4, S7) and on-ear headphones. All tasks were administered individually on the tablets, programmed as Android apps. Standardized instructions were delivered by trained experimenters, and the computerized tasks contained built-in audio instructions presented via headphones. Each task was preceded by a short practice phase to ensure that children understood the instructions.

Across all three time points, children completed the same set of four computerized tasks: (a) a questionnaire assessing attitudes toward mathematics, (b) an executive function task (“Clock and Ball task”), (c) a reading task, and (d) an arithmetic task (see Materials and Measures for details). The mathematics attitudes questionnaire was always administered first. The order of the remaining three tasks was randomized at the class level and kept constant within each class.

As compensation for participation across the three measurement occasions, each class received class money, and the teachers received a book voucher. In addition, children were thanked with a small gift (e.g., a small globe ball, postcard, or paper ball) at the end of the testing sessions.

Tasks and Measures

The present study examined post-error slowing (PES) and post-error accuracy (PEA) in school-related reading and mathematics tasks. In the following, we describe the reading and arithmetic tasks that were used to derive PES and PEA indices and included in the analyses.

Tasks

Reading Task. Reading fluency was assessed with a silent reading task. The task was designed as a single word reading measure suitable for primary school children. Before the main task, children completed a short practice phase to familiarize themselves with the procedure.

On each trial, a written target word was presented above four picture options. Children were instructed to tap the picture that matched the meaning of the word. The three incorrect options served as distractors and were either semantically or phonologically similar to the target. The word and pictures remained on the screen until a response was given. The interstimulus interval was 600 ms, consisting of a 200 ms blank screen, a 200 ms fixation cross, and a subsequent 200 ms blank screen. The assessment comprised 140 items in 2nd grade and 180 items in 4th grade, presented in a fixed order that was identical for all participants. Accuracy and reaction times were recorded on each trial.

Arithmetic Task. The arithmetic task was developed specifically for this study, inspired by the paradigm used by Jacobs et al. (2024). It was implemented as a speeded two-choice task, providing trial-by-trial accuracy and reaction times enabling the computation of post-error slowing indices.

Each child completed 120 arithmetic problems, divided into three blocks: an addition block (36 problems; 18 without and 18 with crossing the tens boundary), a doubling block

(36 problems; 18 without and 18 with crossing the tens boundary), and a mixed block (48 problems) comprising only items below the tens boundary, some newly introduced and some repeated from previous blocks. On each trial, a problem was presented along with two response options—one correct and one incorrect. The incorrect option was designed as a distractor and was generated according to fixed rules (e.g., ±1 or ±10 relative to the correct answer), which were pseudo-randomly distributed across items. Children responded by tapping their chosen answer on a touchscreen.

Before the test blocks, children received standardized spoken instructions via headphones and completed several examples and practice trials with feedback. They were instructed to respond as quickly and accurately as possible. No feedback was provided during the actual test blocks.

Task parameters were adapted to grade level. For 2nd graders, all problems involved single-digit numbers (1–9). For 4th graders, addition problems consisted of a two-digit number plus a single-digit number, and doubling problems involved two-digit numbers; in both cases, results were kept below 100 to maintain age-appropriate difficulty. The presentation time of each arithmetic problem was 4,000 ms for 2nd graders and 3,000 ms for 4th graders, respectively. After this interval, the problem disappeared, but the two response options remained visible until a response was given. The interstimulus interval was 600 ms, consisting of a 200 ms blank screen, a 200 ms fixation cross, and a subsequent 200 ms blank screen.

Post-Error Slowing (PES)

PES was used as an index of monitoring-based strategy adjustments following errors. For each task, we identified trials occurring on the next response after an error (post-error) and, for comparison, trials occurring on the next response after a correct answer (post-correct). PES was then computed as the difference between mean RTs on correct post-error trials and correct post-correct trials, such that positive values indicate slower responding after errors (Schroder et al., 2019):

$$PES = \underline{RT}_{post-error,correct} - \underline{RT}_{post-correct,correct}$$

In the arithmetic task, post-error and post-correct trial pairs were restricted to trials within the same task condition to avoid conflating post-error effects with condition transitions.

Post-Error Accuracy (PEA)

PEA was defined as accuracy on the trial immediately following an error, with post-correct accuracy (PCA) defined analogously for trials following correct responses. For descriptive summaries, PEA and PCA were expressed as percentage correct. To test adaptive control (H4), we modeled post-error accuracy as a function of post-error slowing, expecting greater slowing to predict higher accuracy on the subsequent trial.

Data Screening and Exclusions

To ensure data quality and verify task comprehension, we applied accuracy thresholds based on one-sided exact binomial tests. For the reading task (four response options; chance level = .25), cut-offs differed by version: the 2nd grade version (140 trials) required at least 45 correct responses (32.1%), and the 4th grade version (180 trials) at least 56 correct responses (31.1%). For the arithmetic task (120 trials, two response options; chance level = .50), a minimum of 70 correct responses (58.3%) was required to perform above chance. Cut-offs were derived using the binom.test function in R. Based on these criteria, 2 children were excluded from the reading analyses, and 17 children were excluded from the arithmetic analyses.

Reaction times (RTs) were cleaned to remove implausible values. Trials with RTs below 300 ms were discarded. Upper RT limits were determined through visual inspection of the RT distributions (histograms, boxplots) and were set to 15,000 ms in arithmetic and 20,000 ms in reading. In addition, RTs outside ±3 SD of each participant's mean RT were trimmed (computed on trials within the fixed bounds). In the arithmetic task, participant-specific trimming parameters (mean and SD) were computed separately for each difficulty level. This procedure led to the exclusion of 2.77% (T1) and 2.89% (T2) of reading trials and 4.46% (T1) and 4.25% (T2) of arithmetic trials. Children were required to contribute at least one correct post-error trial at both measurement points to allow post-error calculations.

Children not meeting this criterion were excluded (2 in reading, 8 in arithmetic). In addition, participants with fewer than two error trials were removed to ensure reliable post-error estimates (1 in reading, 14 in arithmetic).

Data Analyses

To test hypotheses 1-3, we used trial-level mixed-effects models separately for the arithmetic and reading tasks to account for the hierarchical data structure (trials nested within children) and to estimate post-error effects without aggregating across trials. This model predicted reaction times from post-error status (post-error vs. post-correct), measurement point (T1 vs. T2), grade (2nd grade vs. 4th grade), and their interactions, with random intercepts and random slopes for post-error effects at the subject level. To test hypothesis 4, we used logistic mixed-effects models with trial-level post-error accuracy (0 = incorrect, 1 = correct) as the dependent variable and z-standardized post-error slowing, measurement point, and grade as fixed effects (random intercepts for children). Statistical significance of fixed effects was evaluated using Type III ANOVA, and significant interactions were followed up with planned post hoc contrasts based on estimated marginal means (emmeans).

Results

Descriptive Statistics

To provide context for the post-error analyses, Table 1 presents descriptive statistics for accuracy, post-error accuracy, post-correct accuracy, reaction time (RT), and post-error slowing (PES) across grades and measurement points in both tasks. Post-error RT and post-correct RT by grade and measurement point are visualized for both tasks in Figure 1.

Table 1

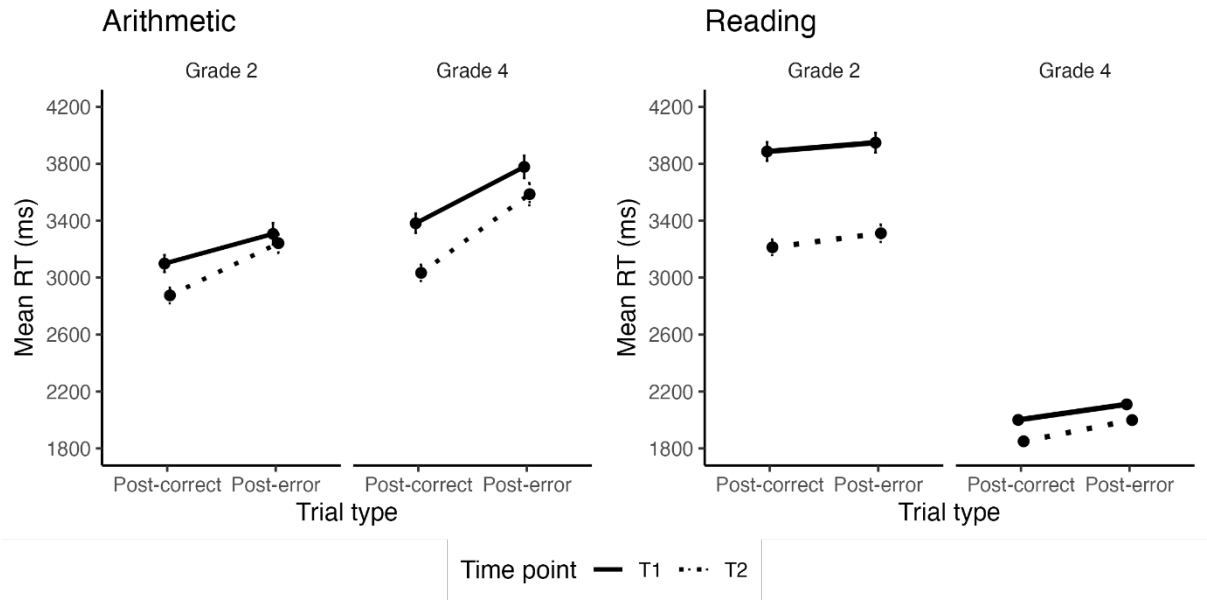
Descriptive Statistics

Grade	Time	Accuracy	Post-error Accuracy	Post-correct Accuracy	RT	PES
Arithmetic						
2	T1	77.27 (12.42)	71.35 (16.51)	79.19 (11.53)	3322.02 (1644.57)	208.57 (757.96)
	T2	85.10 (11.63)	80.83 (16.50)	86.09 (10.75)	3048.74 (1175.84)	366.70 (766.72)
4	T1	77.52 (10.86)	74.92 (13.80)	78.07 (10.84)	3576.21 (1426.26)	396.50 (728.88)
	T2	80.63 (11.22)	77.84 (14.11)	81.15 (10.82)	3181.46 (1245.06)	553.07 (771.95)
Reading						
2	T1	76.90 (16.45)	76.36 (19.24)	77.52 (15.82)	3978.25 (1374.88)	62.17 (524.00)
	T2	83.75 (12.74)	84.17 (16.58)	84.11 (12.04)	3321.70 (1117.92)	98.62 (470.95)
4	T1	86.39 (7.40)	87.93 (10.35)	86.23 (7.09)	2095.70 (552.94)	109.41 (209.32)
	T2	88.13 (7.91)	89.33 (11.94)	88.15 (7.22)	1920.46 (523.41)	149.10 (207.53)

Note. Values are given as Mean (SD). Accuracy is the percentage of correct responses.

Figure 1

Post-correct and Post-error RT by Grade and Measurement Point



Note. Error bars represent standard errors.

Presence of PES (H1)

Consistent with H1, children showed robust post-error slowing (PES) across both tasks and measurement points. In the arithmetic task, reaction times were significantly slower following errors than following correct responses, $F(1, 541) = 106.69$, $p < .001$. The magnitude of PES varied by grade, $F(1, 541) = 17.03$, $p < .001$, and by measurement point, $F(1, 101{,}533) = 58.10$, $p < .001$, as reflected in significant post-error status × grade and post-error status × measurement point interactions. Averaged across grades, PES was significant at both T1 (139 ms) and T2 (362 ms), *p*s < .001. In the reading task, a significant main effect of post-error status when reading likewise indicated reliable PES, $F(1, 598) = 41.43$, $p < .001$. Predicted PES was significant at both T1 (63.8 ms) and T2 (99.2 ms), *p*s < .001, with a modest increase over time reflected in a significant post-error status × measurement point interaction, $F(1, 158{,}361) = 4.16$, $p = .041$, independent of age. Together, these results confirm the presence of PES across tasks and time points, supporting H1, with the magnitude of PES being larger in arithmetic than reading.

Grade Differences in PES (H2)

In line with H2, 4th graders generally exhibited larger PES than 2nd graders, particularly at T1. In the arithmetic task, at T1, PES was significantly larger in 4th graders than in 2nd graders (Δ = 298 ms), $z = 5.40$, $p < .001$. At T2, this grade difference was smaller and no longer statistically significant (Δ = 103 ms), $z = 1.76$, $p = .078$. In the reading task, a similar pattern emerged. Fourth graders showed larger PES than 2nd graders at T1 (Δ = 72 ms), $z = 2.42$, $p = .015$, whereas no significant grade difference was observed at T2 (Δ = 26 ms), $z = 0.81$, $p = .42$. Thus, older children displayed equal or greater post-error slowing than younger children, providing support for H2.

Longitudinal Change in PES (H3)

In the arithmetic task, although overall response times decreased from T1 to T2, $F(1, 100{,}439) = 222.33$, $p < .001$, PES increased over time, as indicated by a significant post-error status × measurement point interaction, $F(1, 101{,}533) = 58.10$, $p < .001$. Importantly, this increase was larger for 2nd graders (Δ = 321 ms), $z = 7.54$, $p < .001$, than for 4th graders

(Δ = 126 ms), $z = 3.13$, $p = .002$, reflected in a significant post-error status × measurement point × grade interaction, $F(1, 101{,}533) = 11.04$, $p < .001$. The difference in change between grades was significant (Δ = −195 ms), $z = -3.32$, $p < .001$. In the reading task, PES increased modestly from T1 to T2, $F(1, 158{,}361) = 4.16$, $p = .041$, with no evidence that this change was moderated by grade (post-error status × measurement point × grade: $F(1, 158{,}251) = 1.82$, $p = .177$). Taken together, there were substantial longitudinal changes in PES over the school year confirming H3, with these changes differing by grade in the arithmetic task, but were comparable across grades in the reading task, suggesting task-specific developmental changes of post-error adjustments.

PES and PEA Link (H4)

In the arithmetic task, the effect of post-error slowing on post-error accuracy differed across grade and measurement point, as indicated by a significant three-way interaction (slowing × measurement point × grade, β = 0.20, SE = 0.06, $z = 3.19$, $p = .001$). Simple slopes analyses revealed that for 2nd graders, greater post-error slowing was associated with higher post-error accuracy at both T1 (β = 0.21, SE = 0.03, 95% CI [0.16, 0.27]) and T2 (β = 0.09, SE = 0.04, 95% CI [0.02, 0.16]). This corresponds roughly to a 24% increase in the odds of post-error accuracy per one standard deviation increase in slowing at T1, decreasing to about 9% at T2 (slope difference T1 vs. T2 for grade 2: β = −0.12, SE = 0.04, $z = -2.80$, $p = .005$). For 4th graders, the effect of PES on PEA was not significant at T1 (β = 0.02, SE = 0.03, 95% CI [−0.03, 0.08], ≈2% increase in odds) and modestly positive at T2 (β = 0.10, SE = 0.03, 95% CI [0.03, 0.16], ≈10% increase in odds), with a non-significant increase in slope from T1 to T2 (β = 0.07, SE = 0.04, $z = 1.70$, $p = .090$). In addition, post-error accuracy was higher at T2 than at T1 (β = 0.35, SE = 0.04, $z = 7.95$, $p < .001$, OR ≈ 1.42) and slightly higher in 4th than in 2nd graders (β = 0.20, SE = 0.06, $z = 3.34$, $p < .001$, OR ≈ 1.22).

In the reading task, the effect of PES on PEA differed across grade and measurement point, as indicated by a significant three-way interaction (slowing × measurement point × grade, β = 0.33, SE = 0.09, $z = 3.63$, $p < .001$). Simple slopes analyses revealed that for 2nd graders, greater post-error slowing was associated with lower

post-error accuracy at both T1 (β = −0.09, SE = 0.02, 95% CI [−0.12, −0.05]) and T2 (β = −0.08, SE = 0.03, 95% CI [−0.13, −0.03]). This corresponds roughly to an 8–9% decrease in the odds of post-error accuracy per one standard deviation increase in post-error slowing. For 4th graders, slowing had a strong negative effect at T1 (β = −0.44, SE = 0.05, 95% CI [−0.55, −0.33], ≈36% decrease in odds) but was attenuated at T2 (β = −0.10, SE = 0.07, 95% CI [−0.23, 0.03], ≈9% decrease in odds), with a significant increase in slope from T1 to T2 (β = 0.34, SE = 0.09, z = 3.98, p < .001). In addition, post-error accuracy was higher at T2 than at T1 (β = 0.56, SE = 0.04, z = 14.26, p < .001, OR ≈ 1.75) and higher in 4th than in 2nd graders (β = 0.90, SE = 0.08, z = 10.63, p < .001, OR ≈ 2.46).

Discussion

The present study examined children's balancing of speed and accuracy demands, specifically their post-error adjustments, in two academic domains—reading and arithmetic—under ecologically valid, yet experimentally controlled conditions, combining a cross-sectional (comparing 2nd and 4th graders) with a short-term longitudinal design. We aimed to explore whether post-error slowing (PES) emerges reliably in curriculum-based tasks, how it develops over short periods of time in middle childhood, and whether it reflects efficient strategy use in terms of performance benefits. Overall, the findings provide strong evidence that children robustly slow down after errors in both academic domains, but they also reveal pronounced task- and age-related differences in the magnitude of PES, its developmental trajectories, and functional significance which we will discuss in the following paragraphs.

Robust Presence of PES Across Tasks and Time

Consistent with our first hypothesis and in line with de Mooij et al. (2022), children showed clear and reliable post-error slowing in both arithmetic and reading at both measurement points. These findings extend prior work demonstrating substantial PES in children using cognitive conflict paradigms (Dubravac et al., 2022; Ger & Roebers, 2023a, 2023b; Gupta et al., 2009) to curriculum-based mathematical and reading tasks administered in educational settings. While often overlooked, elementary school children

were found to detect their mistakes (substantial error monitoring; Botvinick et al., 2001; Braver, 2012; Roebers, 2017) and to strategically and specifically upregulate their cognitive control processes. The current findings provide strong support for the generality of PES in childhood and establish its presence in foundational academic skills – domains in which research on children's strategy use has neglected the balancing of speed and accuracy.

Similar to the results reported by de Mooij and colleagues (2022), PES magnitude was larger for arithmetic than for reading (see Figure 1). We are tempted to rule out task difficulty's responsibility for the PES differences across reading and arithmetic because performance was very similar (see Table 1); nevertheless and admittedly, even small variations of item difficulty can alter PES – which might have been more the case in our arithmetic than reading task (see also de Mooij et al., 2022). Accurate and fast word reading followed by selecting one of four alternatives may also be structurally different from arithmetic fact retrieval (with a two-choice answering format) and therefore be less a matter trading off speed against accuracy. In other words, the difference in PES magnitude may be due to distinct processing demands across domains and calls for further investigation.

Cross-sectional Age Differences in PES Magnitude

In our study, 4th graders showed equal or larger post-error slowing than 2nd graders, particularly at the first measurement. This pattern was observed in both arithmetic and reading, with grade differences diminishing or disappearing by the end of the school year. Thus, older children in our sample appeared at least as sensitive—and in some cases more sensitive—to errors than younger children and this is in line with developmental progression in metacognitive monitoring, strategy use, and the accuracy-speed trade-off (Bellon et al., 2019; Brod et al., 2020; Camerota et al., 2019; Krebs & Roebers, 2010; for a review see Roebers, 2022b). As we assume that the underlying arithmetic and reading operations are not yet fully automated in this age, children still needed to invest substantial cognitive effort. To perform successfully, thus, they may continue to engage in substantial PES, a pattern consistent with evidence that PES is larger under high task demands (de Mooij et al., 2022; Ger & Roebers, 2023a, 2024; Cravet & Ger, 2026).

Longitudinal Developmental Change in PES

Our third hypothesis concerned longitudinal change in PES. The results provide a complex, task-dependent picture: In arithmetic, PES increased from the first to the second measurement despite overall reductions in response time and improvement in accuracy, indicating that post-error slowing became more pronounced as children gained more experience with the speeded task and basic arithmetic. Crucially, this increase in PES was significantly larger in 2nd graders than in 4th graders. This age-dependent increase is consistent with the notion that monitoring and strategy use rapidly advance in the early elementary school years (Schneider & Löffler, 2016) and also aligns with developmental perspectives suggesting that children learn to use error signals more effectively in this age range before PES can eventually decline with further automatization (de Mooij et al., 2022; Gupta et al., 2009). Clearly, more longitudinal data is necessary to fully understand developmental changes in arithmetic's PES.

In contrast, reading showed only a modest increase in PES over time, with no evidence of grade moderation. This divergence between domains suggests that developmental changes in strategic post-error adjustments are not uniform across academic skills. Arithmetic fact retrieval may place greater demands on monitoring and control processes during this developmental window, whereas reading fluency—particularly in transparent orthographies or highly practiced contexts—may rely more heavily on automatized processes that are less sensitive to error detection and thus PES. Importantly, these domain-specific findings indicate that developmental change can be detected when tasks are sufficiently sensitive, reaction time is measured at the trial level, and multiple age cohorts are included.

Functional Significance of PES

The final goal of the study was to test whether PES reflects efficient strategy use and adaptive control by examining its association with post-error accuracy (PEA). Here, the results revealed a striking dissociation between domains: In arithmetic, greater post-error slowing was associated with higher post-error accuracy. This effect was strongest in younger

children at T1, with a 24% increase in the odds of correct responses per one standard deviation increase in slowing and diminished over time (≈9% increase at T2). For older children, the benefit of slowing was smaller (2–10% increase in odds). These results support the view of strategic and efficient speed and accuracy balancing in children's arithmetic, enhancing subsequent performance (Bjorklund et al., 1997; Botvinick et al., 2001; Laming, 1968). The effect size was modest but reliable, aligning with prior evidence linking PES to improved performance in mathematical tasks (de Mooij et al., 2022; Jacobs et al., 2024). Thus, in arithmetic, PES can be considered as a beneficial strategy, consistent with the cognitive control view that errors trigger compensatory increases in control that help prevent error repetition. Moreover, our results suggest that younger children rely more on slowing to improve accuracy, whereas older children increase their speed of responding maintaining or even increasing their accuracy and then, PES may mirror more a disruption of their processing.

In reading, the pattern was qualitatively different. Greater post-error slowing predicted lower post-error accuracy, particularly in 4th graders at T1 (≈36% decrease in odds) and to a smaller extent in 2nd graders (≈8–9% decrease in odds). By T2, the negative effect of slowing was attenuated for both grades. This overall negative association suggests that slowing after errors in reading does not reflect an effective strategy and may instead index disruption, uncertainty, or inefficient processing. Such a pattern is consistent with the view that slowing may reflect attentional capturing by unexpected events rather than strategic control (Notebaert et al., 2009). In reading, where fluent performance depends on rapid lexical access and continuous processing, interrupting the flow after an error may be detrimental rather than beneficial. Future studies may need to systematically vary the task demands to circumscribe under which conditions slowing down (after an error or when comprehension difficulties arise) might still be a valuable strategy.

Notably, the stronger positive PES–PEA association in arithmetic (and the smaller negative association in reading) in younger children, despite older children showing generally greater PES, may indicate a developmental shift in the functional role of post-error

slowing. In early childhood, PES appears more adaptive, supporting performance improvement after errors, whereas in older children, PES may increasingly reflect orienting or attentional capturing, particularly when slowing no longer translates into better accuracy. This pattern suggests that the meaning and utility of PES may change over development, with strategic benefits declining as task skills become more automated.

Limitations

Several limitations of the present study should be considered. First, although the arithmetic and reading tasks were age- and curriculum-appropriate, they captured only specific and very constrained aspects of children's academic performance. Second, the study focused on two discrete grades (2nd and 4th), which may limit the precision with which developmental trajectories of PES across middle childhood can be characterized. Finally, although we accounted for grade and measurement point, individual differences in cognitive and metacognitive skills, motivation, or prior experience with speeded arithmetic and reading were not directly controlled, which may contribute to variability in PES and post-error accuracy. Future research should examine PES across a broader range of grades, tasks, and individual characteristics to more fully understand its strategic role in academic development.

Implications and Conclusions

Taken together, the present findings demonstrate that post-error slowing is a robust yet often overlooked marker of children's implicit and strategic error-related adjustments in both reading and arithmetic. PES was found to increase over time in arithmetic—especially among younger children—and was associated with improved post-error accuracy, consistent with interpretations surrounding strategy use and improving cognitive control upregulations. In reading, however, PES showed weaker developmental modulation and was negatively related to post-error accuracy, suggesting that slowing in reading may reflect maladaptive interruptions of ongoing reading processes rather than beneficial regulations.

The present study helps bridge the gap between laboratory-based research and real-world educational contexts. We found clear evidence that 2nd graders monitor their

performance and strategically upregulate their cognitive control by selectively slowing down after errors to the benefit of their arithmetic performance. Hence, errors in arithmetic should be considered and treated as productive learning opportunities, not as failures. Coming back to Susie in our introductory example, Susie and her classmates might need more time for the test, to allow a good balancing of speed and accuracy and to foster error monitoring and error correction. Moreover, arithmetic instructional practices should provide reflection prompts, motivate re-checking, and foster balancing speed and accuracy. Error handling in reading, in contrast, may be better addressed outside speeded contexts, for example, through feedback and guided re-reading. Our findings also point to the possibility that educational, classroom-based and timed assessment methods, especially in arithmetic, likely conflate domain-specific skills with monitoring and control processes, and performance might not yield the most valid and a pure picture of children's domain-specific learning progress.

Given PES's theoretical and practical relevance, future work should investigate the cognitive and metacognitive mechanisms underlying these domain-specific patterns of PES, strategy use and its efficiency. A promising direction for future research is to examine whether specific instructions, prompts or feedback can shape the adaptiveness of children's post-error adjustments, as suggested by recent training work showing feedback-related improvements in PES (Ger & Roebers, 2024b).

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